

Gurunanak Adhikari

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Professional Summary

Cybersecurity student and ISC2 CC with technical exposure to Blue Team and Red Team fundamentals through structured internships, labs, and projects. Experienced in AWS cloud, basic web application testing, network traffic analysis, SIEM fundamentals, CTF challenges, and Python-based security scripting. Seeking opportunities in cybersecurity.

Education

IIMS College (Affiliated with Taylor's University) <i>Bachelor of Computer Science (Specialization in Cybersecurity)</i>	Kathmandu, Nepal 2023 – 2027 (Expected)
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Professional Experience

FlyRank AI <i>Backend AI Engineer Intern (Part-time)</i>	Remote Jul 2026 – Sept 2026
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- Built Node.js/Express REST APIs for task management using repository patterns.
- Containerized PostgreSQL database and application stack via Docker Compose.
- Implemented Supabase JWT authentication, protected routes, and middleware.
- Developed web scraping pipeline with caching, throttling, and Zod validation.
- Designed and built *Google Forms Scam Analyzer*, a Chrome extension + FastAPI backend capstone that classifies scam risk via LLM tactic analysis (Groq/Gemini fallback) and concurrent OSINT reputation checks (Google Safe Browsing, VirusTotal, urlscan.io, Tavily), deployed live with a bring-your-own-key architecture and per-IP rate limiting.

Cloudmandap <i>Data Automation and AI Fellow (AWS DevOps)</i>	Kathmandu, Nepal Feb 2026 – Aug 2026
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- Deployed, configured, and managed lab-scale AWS infrastructure using core services, DevOps workflows, and security-focused automation principles.
- Completed curriculum tracks for AWS Cloud Practitioner and Solutions Architect Associate pathways.

Guided Virtual Internships <i>Cyber Security Extern (ShadowFox, Redynox, HACK SECURE)</i>	Remote Apr 2025 – Aug 2025
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- **Traffic Analysis & Reconnaissance:** Inspected packet flow and protocol behavior via Wireshark; executed subdomain discovery, DNS analysis, and web asset gathering.
- **Vulnerability Assessment & Hardening:** Evaluated system logs for anomalies and assessed open ports/firewall rules within guided sandbox environments.
- **Application Security Foundations:** Solved foundational web security CTFs (SQLi/XSS) and built Python security scripts for sniffing and password verification.

Technical Labs & Projects

Multi-Node Global Threat Intelligence Pipeline & Open Dataset	GitHub
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- Built and operated a 3-node CTI pipeline on DigitalOcean across 3 months, capturing, deduplicating, and openly publishing a 47.6K-log threat dataset (Zenodo DOI) and reporting 1,755 high-severity threat IPs to AbuseIPDB.
- Engineered an automated GitHub Actions backend using Vertex AI Gemini API for payload classification; audited LLM severity labels against human review ($n = 100$), improving inter-rater reliability from Cohen's $\kappa = 0.66$ to 0.80 by correcting systematic labeling inconsistencies.
- Enforced Principle of Least Privilege via Linux kernel capabilities (`cap_net_bind_service`) and visualized real-time telemetry via an interactive 3D threat globe (Three.js/Tailwind CSS).

TraceWrite – Secure Collaborative Platform & AppSec Audit	tracewrite.fasttype.vip
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- **GCP Deployment & Hosting:** Built and hosted application on a Google Cloud Platform (GCP) Compute Engine VM configured with Nginx reverse proxy and custom domain DNS routing.
- **Testing & DAST Security Scan:** Executed Dynamic Application Security Testing (DAST) using OWASP ZAP and built a 120+ test suite (including 50 automated unit tests) to validate platform security and functional logic.
- **Vulnerability Remediation:** Identified and patched a critical Insecure Direct Object Reference (IDOR/RBAC) vulnerability to prevent unauthorized data downloads, and updated vulnerable legacy dependencies (html2pdf).

- **Architecture & BYOK Security:** Built a Manifest V3 Chrome extension and FastAPI backend where all LLM/OSINT API keys are supplied per-user (never stored server-side), transmitted over TLS as request headers, with per-IP rate limiting protecting shared demo keys from abuse.
- **Multi-Provider LLM Pipeline:** Implemented structured-output classification (Groq `gpt-oss-120b` primary, Gemini 3.6 Flash fallback) for scam-tactic detection and named-entity extraction, with provider order determined by live reliability testing rather than initial assumptions.
- **Concurrent OSINT Integration:** Engineered an async background-job pipeline (FastAPI `BackgroundTasks`) running four reputation APIs (Safe Browsing, VirusTotal, urlscan.io, Tavily) in parallel via `ThreadPoolExecutor`, decoupled from the synchronous request path.
- **False-Positive Remediation:** Diagnosed and fixed coincidental keyword-match false positives in entity reputation checks by redesigning the relevance judgment to require quoted, sourced evidence before flagging any entity.
- Deployed live on Render with a Postgres-ready SQLAlchemy data layer and content-hash caching to avoid redundant API calls.

AWS SOAR Engine

GitHub

- Built an automated serverless SOAR pipeline using AWS WAF and ALB to block Layer 7 exploits on a vulnerable DVWA host. Upon attack detection, a custom Python Lambda function parses S3 threat logs, dynamically swaps the target EC2 instance into a zero-ingress Quarantine security group, and uses the Twilio Voice API to trigger out-of-band emergency calls.

AWS & Splunk SOC Lab

GitHub

- Built a distributed cloud SOC lab by deploying an isolated instance hosting a containerized DVWA application via Docker. Configured a Splunk Universal Forwarder to route JSON container logs to Splunk Enterprise, validating log ingestion telemetry via simulated directory traversal and path attacks.

Technical Skills

- **Tools:** OWASP ZAP, Netlify, GCP Compute Engine, Wireshark, Nmap, SQLmap, Snort, Splunk, Kali Linux, Docker, Nginx.
- **Concepts:** Application Security (AppSec), DAST, Cloud Security Architecture (AWS/GCP), Incident Automation (SOAR), Log Analysis & SIEM, Network Security, Threat Hunting.
- **Programming & AI Tools:** Basic Python for security scripting, JavaScript/HTML/CSS, Lovable AI, GitHub Copilot, Cline, Cursor

Certifications & Professional Development

Professional Certifications

- ISC2 Certified in Cybersecurity (CC)
- AWS Cloud Practitioner

Coursework & Technical Training

- **Coursera:** Splunk Knowledge Manager & Search Expert
- **AWS Academy:** Solutions Architect & Cloud Practitioner
- **Google, IBM:** Cybersecurity Professional Certificate